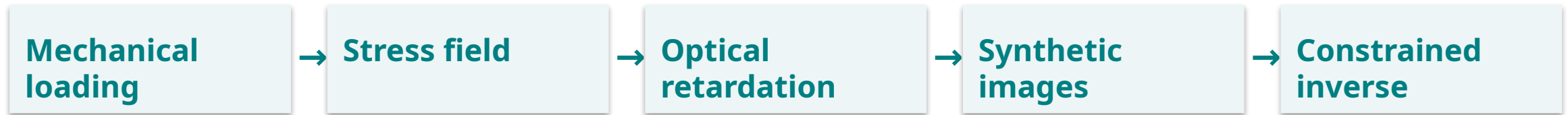


From mechanical stress to polarized-light images

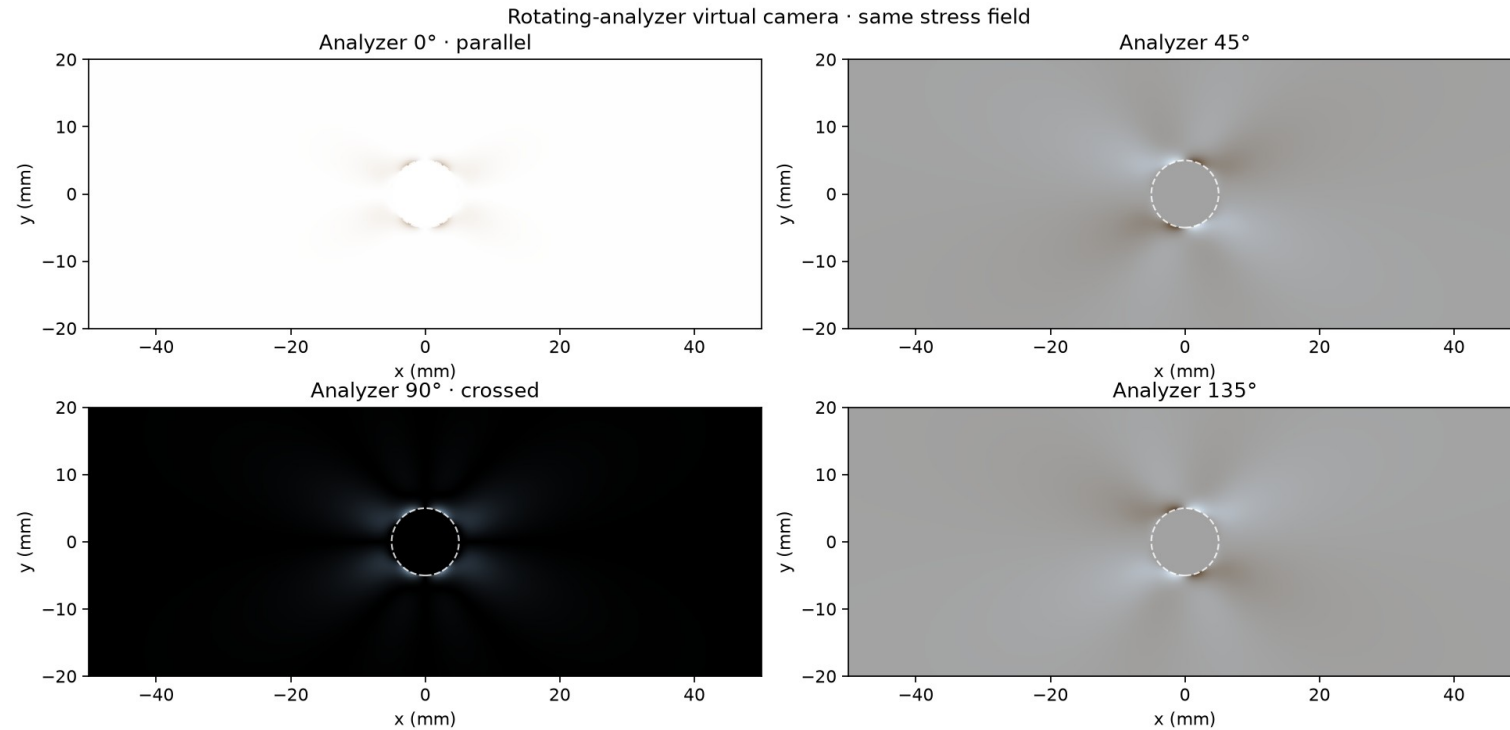
Objective

Predict the optical response around a circular hole and explore where stress differences can be recovered from synthetic images.



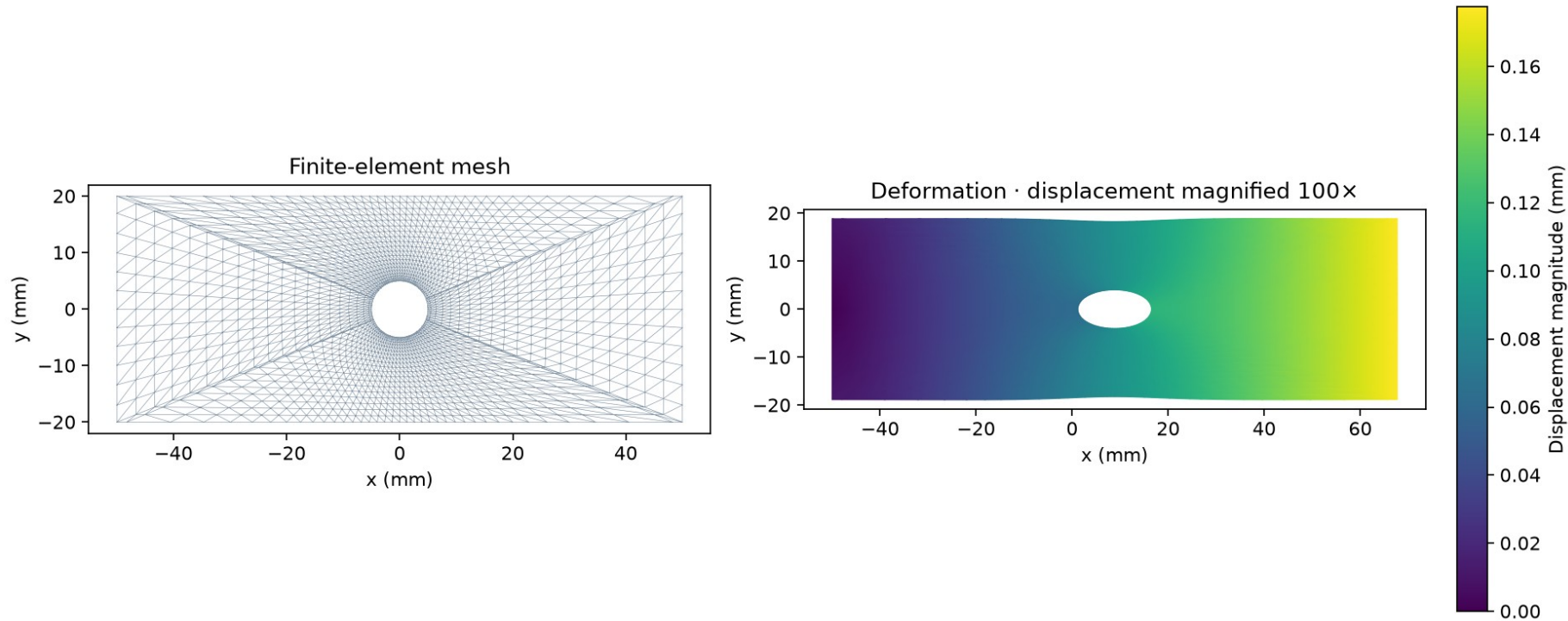
- Mechanics: analytical and finite-element models provide stress differences and principal-axis orientations.
- Optics: the stress-optic law and Jones model predict wavelength- and analyzer-dependent intensity.
- Inverse: recover stress differences only where measurements are informative, with explicit uncertainty.

Image 1 — Effect of analyzer rotation



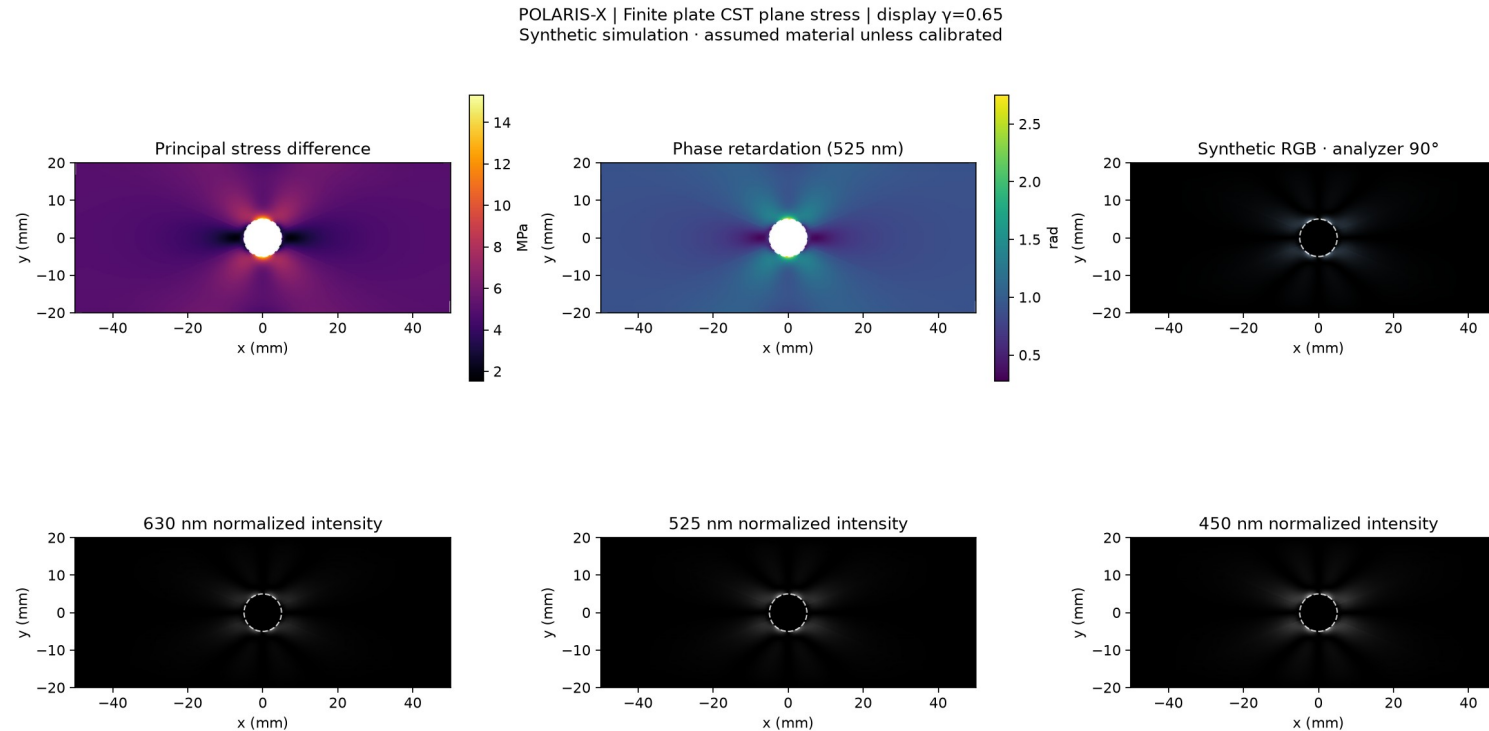
- Same stress field viewed at analyzer angles of 0°, 45°, 90° and 135°.
- Parallel polarizers give a bright background; crossed polarizers reveal localized optical contrast.
- Image brightness depends on optical orientation and retardation, as well as the display gamma.

Image 2 — Finite-element mesh and deformation



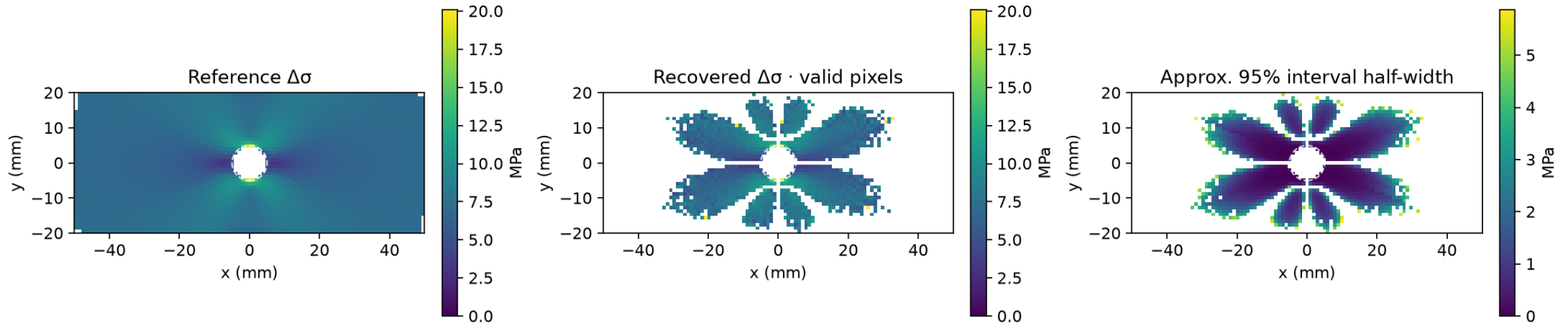
- Triangular elements are refined near the hole, where stress changes rapidly.
- The deformed shape is magnified 100x to make small displacements visible.
- The color scale gives actual displacement magnitude in mm; its values are not magnified.

Image 3 — Finite-plate stress-to-light simulation



- Principal stress difference is elevated near the upper and lower hole boundary.
- The stress-optic law converts stress difference to retardation; Jones optics predicts the intensity.
- Synthetic RGB combines 630, 525 and 450 nm channels, shown separately below.

Image 4 — Stress reconstruction and uncertainty



- Compare reference stress difference with recovered values at valid pixels.
- Blank reconstruction regions are unresolved or outside material; they do not mean zero stress.
- The approximate 95% interval half-width quantifies local uncertainty in MPa.